



INSTITUTO POLITÉCNICO NACIONAL

ESCUELA SUPERIOR DE FÍSICA Y MATEMÁTICAS

CATALOGO DE DISTRIBUCIONES DE
PROBABILIDAD

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GRUPO: 7MM1

MATERIA: INGENIERIA INDUSTRIAL

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1. Distribuciones Discretas

Bernoulli

$$X \sim \text{Bern}(p)$$

Parámetros $0 \leq p \leq 1.$

Soporte $x \in \{0, 1\}.$

PMF $p_X(x)(x) = p^x(1-p)^{1-x}.$

CDF

$$F_X(x)(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

Media $\mathbb{E}[X] = p.$

Varianza $\text{Var}(X) = p(1-p).$

Desv. est. $\text{DE}(X) = \sqrt{p(1-p)}.$

MGF $M_X(t) = (1-p) + pe^t.$

Binomial

$$X \sim \text{Bin}(n, p)$$

Parámetros $n \in \mathbb{N}, 0 \leq p \leq 1.$

Soporte $x = 0, 1, 2, \dots, n.$

PMF $p_X(x)(x) = \binom{n}{x} p^x (1-p)^{n-x}.$

CDF

$$F_X(x)(x) = \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k (1-p)^{n-k}$$

Media $\mathbb{E}[X] = np.$

Varianza $\text{Var}(X) = np(1-p).$

Desv. est. $\text{DE}(X) = \sqrt{np(1-p)}.$

MGF $M_X(t) = ((1-p) + pe^t)^n.$

Parámetros	$0 < p \leq 1.$
Soporte	$x = 1, 2, 3, \dots$
PMF	$p_X(x)(x) = (1 - p)^{x-1}p.$
CDF	$F_X(x)(x) = 1 - (1 - p)^{\lfloor x \rfloor}, \text{ para } x \geq 1.$
Media	$\mathbb{E}[X] = \frac{1}{p}.$
Varianza	$\text{Var}(X) = \frac{1-p}{p^2}.$
Desv. est.	$\text{DE}(X) = \frac{\sqrt{1-p}}{p}.$
MGF	

$$M_X(t) = \frac{pe^t}{1 - (1 - p)e^t}, \quad t < -\ln(1 - p)$$

Parámetros	$r \in \mathbb{N}, 0 < p \leq 1.$
Soporte	$x = r, r + 1, r + 2, \dots$
PMF	$p_X(x)(x) = \binom{x-1}{r-1} p^r (1 - p)^{x-r}.$
CDF	
	$F_X(x)(x) = \sum_{k=r}^{\lfloor x \rfloor} \binom{k-1}{r-1} p^r (1 - p)^{k-r}$
Media	$\mathbb{E}[X] = \frac{r}{p}.$
Varianza	$\text{Var}(X) = \frac{r(1-p)}{p^2}.$
Desv. est.	$\text{DE}(X) = \frac{\sqrt{r(1-p)}}{p}.$
MGF	

$$M_X(t) = \left(\frac{pe^t}{1 - (1 - p)e^t} \right)^r, \quad t < -\ln(1 - p)$$

Parámetros $\lambda > 0.$

Soporte $x = 0, 1, 2, \dots$

PMF $p_X(x)(x) = e^{-\lambda} \frac{\lambda^x}{x!}.$

CDF

$$F_X(x)(x) = \sum_{k=0}^{\lfloor x \rfloor} e^{-\lambda} \frac{\lambda^k}{k!}$$

Media $\mathbb{E}[X] = \lambda.$

Varianza $\text{Var}(X) = \lambda.$

Desv. est. $\text{DE}(X) = \sqrt{\lambda}.$

MGF $M_X(t) = \exp(\lambda(e^t - 1)).$

Parámetros $a, b \in \mathbb{Z}, a < b.$ Sea $n = b - a + 1.$

Soporte $x \in \{a, a+1, \dots, b\}.$

PMF $p_X(x)(x) = \frac{1}{n}.$

CDF

$$F_X(x)(x) = \begin{cases} 0, & x < a \\ \frac{\lfloor x \rfloor - a + 1}{n}, & a \leq x < b \\ 1, & x \geq b \end{cases}$$

Media $\mathbb{E}[X] = \frac{a+b}{2}.$

Varianza $\text{Var}(X) = \frac{n^2 - 1}{12}.$

Desv. est. $\text{DE}(X) = \sqrt{\frac{n^2 - 1}{12}}.$

MGF

$$M_X(t) = \frac{1}{n} \sum_{k=0}^{n-1} e^{t(a+k)} = \frac{e^{ta}(e^{tn} - 1)}{n(e^t - 1)}, \quad t \neq 0; \quad M_X(t)(0) = 1$$

2. Distribuciones Continuas

Uniforme Continua

$X \sim \text{Unif}(a, b)$

Parámetros	$a < b.$
Soporte	$a \leq x \leq b.$
PDF	$f_X(x)(x) = \frac{1}{b-a}, \text{ para } a \leq x \leq b.$
CDF	$F_X(x)(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$
Media	$\mathbb{E}[X] = \frac{a+b}{2}.$
Varianza	$\text{Var}(X) = \frac{(b-a)^2}{12}.$
Desv. est.	$\text{DE}(X) = \frac{b-a}{\sqrt{12}}.$
MGF	$M_X(t) = \frac{e^{tb} - e^{ta}}{t(b-a)}, \quad t \neq 0; \quad M_X(t)(0) = 1$

Exponencial

$X \sim \text{Exp}(\lambda)$

Parámetros	$\lambda > 0.$
Soporte	$x \geq 0.$
PDF	$f_X(x)(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$
CDF	$F_X(x)(x) = 1 - e^{-\lambda x}, \quad x \geq 0.$
Media	$\mathbb{E}[X] = \frac{1}{\lambda}.$
Varianza	$\text{Var}(X) = \frac{1}{\lambda^2}.$
Desv. est.	$\text{DE}(X) = \frac{1}{\lambda}.$
MGF	$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$

Parámetros $\alpha > 0$ (forma), $\lambda > 0$ (tasa).

Soporte $x \geq 0$.

PDF

$$f_X(x)(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x \geq 0$$

CDF $F_X(x)(x) = \frac{\gamma(\alpha, \lambda x)}{\Gamma(\alpha)}$ (gamma incompleta).

Media $\mathbb{E}[X] = \frac{\alpha}{\lambda}$.

Varianza $\text{Var}(X) = \frac{\alpha}{\lambda^2}$.

Desv. est. $\text{DE}(X) = \frac{\sqrt{\alpha}}{\lambda}$.

MGF $M_X(t) = \left(\frac{\lambda}{\lambda - t} \right)^\alpha, \quad t < \lambda$.

Parámetros $\mu \in \mathbb{R}, \sigma > 0$.

Soporte $x \in \mathbb{R}$.

PDF

$$f_X(x)(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

CDF $F_X(x)(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$ (no tiene forma elemental cerrada).

Media $\mathbb{E}[X] = \mu$.

Varianza $\text{Var}(X) = \sigma^2$.

Desv. est. $\text{DE}(X) = \sigma$.

MGF $M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$.

Definición Si $Y \sim \mathcal{N}(\mu, \sigma^2)$, entonces $X = e^Y$ es lognormal.

Soporte $x > 0$.

PDF

$$f_X(x)(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$$

CDF

$$F_X(x)(x) = \Phi\left(\frac{\ln x - \mu}{\sigma}\right), x > 0.$$

Media

$$\mathbb{E}[X] = \exp\left(\mu + \frac{\sigma^2}{2}\right).$$

Varianza

$$\text{Var}(X) = (e^{\sigma^2} - 1) e^{2\mu + \sigma^2}.$$

Desv. est.

$$\text{DE}(X) = \sqrt{(e^{\sigma^2} - 1) e^{2\mu + \sigma^2}}$$

MGF

No existe para $t > 0$ (diverge). Para $t \leq 0$ puede definirse como integral, pero no tiene forma cerrada simple.

Parámetros $\nu > 0$ (grados de libertad).

Relación $X \sim \Gamma(\alpha = \nu/2, \lambda = 1/2)$.

Soporte $x \geq 0$.

PDF

$$f_X(x)(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}$$

CDF

$$F_X(x)(x) = \frac{\gamma(\nu/2, x/2)}{\Gamma(\nu/2)}.$$

Media

$$\mathbb{E}[X] = \nu.$$

Varianza

$$\text{Var}(X) = 2\nu.$$

Desv. est.

$$\text{DE}(X) = \sqrt{2\nu}.$$

MGF

$$M_X(t) = (1 - 2t)^{-\nu/2}, \quad t < \frac{1}{2}.$$

Parámetros $\mu \in \mathbb{R}, b > 0.$

Soporte $x \in \mathbb{R}.$

PDF

$$f_X(x)(x) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right)$$

CDF

$$F_X(x)(x) = \begin{cases} \frac{1}{2} \exp\left(\frac{x - \mu}{b}\right), & x < \mu \\ 1 - \frac{1}{2} \exp\left(-\frac{x - \mu}{b}\right), & x \geq \mu \end{cases}$$

Media $\mathbb{E}[X] = \mu.$

Varianza $\text{Var}(X) = 2b^2.$

Desv. est. $\text{DE}(X) = \sqrt{2}b.$

MGF $M_X(t) = \frac{e^{\mu t}}{1 - b^2 t^2}, \quad |t| < \frac{1}{b}.$